

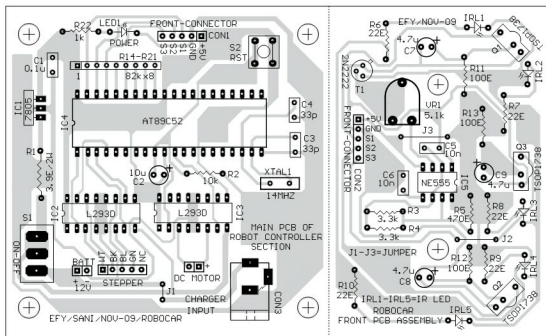


Fig. 5: Component layout for the PCB

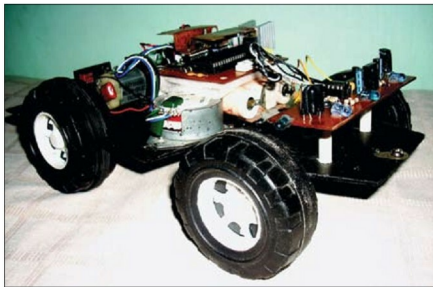


Fig. 6: Author's prototype of RoboCar

The circuit is assembled on a general-purpose PCB or on a PCB layout. The actual size, single-side PCB layout is shown in Fig. 4 and its component layout in Fig. 5.

The main PCB and the front PCB can be separated by cutting along the dotted line shown in the PCB layout.

The front PCB assembly is mounted on the front side of the robot as shown in Fig. 3. It is attached firmly on the

main chassis board with nuts, bolts and spacers. The main PCB is also mounted on the chassis board with nuts and bolts and spacers. Three 4V rechargeable battery cells, taken from a laptop, can be placed between the main PCB and the chassis board (Fig. 6). The 'charger input' charging terminal has been provided in the PCB for charging the battery.

The main PCB and the front PCB are connected through CON1 and

CON2 with suitable length of wires.

Software

The software code is written in C language using the free small device C compiler or SDCC. You can download SDCC from the link <http://sourceforge.net/projects/sdcc/files/>, free of cost. The compiler creates some issues in Windows XP. In this project the code is compiled in a Windows 98 SE environment. The generated hex code is then burnt into

the microcontroller unit (MCU) using a suitable Atmel 89 series programmer such as one from Sunrom Technologies or Frontline Electronics. You should not remove the microcontroller from the zero insertion force (ZIF) socket until the programming is complete.

Steps for installation

1. Install SDCC using SDCC-2.9.0-setup file. It will automatically install under c:\Program Files\SDCC
2. Copy 'Robo1.c' file under C:\Program Files\SDCC\Robocar
3. Open the MSDOS prompt and give the above path
4. Type 'SDCC Robo1.c' against the DOS prompt to compile it. If no errors are found in the code, it will generate the .lst and .ihx files in the same directory where the robo1.c file is located
5. Convert the Robo1.ihx file to generate the robo1.hex file using 'pack.ihx' as:

```
packihx robo1.ihx>robo1.hex.
```

6. Use the robo1.hex file to program the MCU

Precaution

Once all the parts are assembled, it is necessary to check the operational